



7 – Discrete. Bivariate Dist. (again)

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Recap: Conditional PDF

- The conditional PMF/PDF of X , given that $Y = y$, is defined by

$$g(x|y) = \frac{f(x, y)}{f_Y(y)}.$$

- The conditional PMF/PDF of Y , given that $X = x$, is defined by

$$h(y|x) = \frac{f(x, y)}{f_X(x)}.$$

*Relevant for both
Continuous and Discrete

Recap: Conditional Mean/Variance

- Conditional mean of Y given $X = x$:

$$\mu_{Y|x} = \mathbb{E}[Y|X = x] = \sum_{y \in Y} yh(y|x)$$

- Conditional variance of Y given $X = x$:

$$\sigma_{Y|x}^2 = \mathbb{E}[Y^2|X = x] - (\mu_{Y|x})^2$$

$= \sum_{y \in Y} y^2 h(y|x)$

Recap: Easy mix-up

- The conditional PMF/PDF of Y , given that $X = x$, is defined by

$$h(y|x) = \frac{f(x, y)}{f_X(x)}.$$

	$X < 5$	$X = 5$	$X > 5$
$Y = 0$	...	1/10	...
$Y = 1$	...	2/10	...
$Y = 2$	...	1/10	...
	...	$f_X(X=5) = 4/10$	...

$$h(y|x = 5) \neq h(y, x = 5)$$

Problem 1 (Joint PMF, Marginal PMF, Conditional PMF)

Let W equal the weight of laundry soap in a 1-kilogram box that is distributed in Southeast Asia. Suppose that $\mathbb{P}(W < 1) = 0.02$ and $\mathbb{P}(W > 1.072) = 0.08$. Call a box of soap light, good, or heavy depending on whether $W < 1$, $1 \leq W \leq 1.072$, or $W > 1.072$, respectively. In $n = 50$ independent observations of these boxes, let X equal the number of light boxes and Y the number of good boxes.

- (a) What is the joint PMF of X and Y ?
- (b) Give the name of the distribution of Y along with the values of the parameters of this distribution.
- (c) Given that $X = 3$, how is Y distributed conditionally?
- (d) Determine $\mathbb{E}[Y|X = 3]$

Problem 2 (Joint PMF, Marginal PMF, Conditional PMF)

An insurance company sells both homeowners' insurance and automobile deductible insurance. Let X be the deductible on the homeowners' insurance and Y the deductible on automobile insurance. Among those who take both types of insurance with this company, we find the following probabilities:

	$x = 100$	$x = 500$	$x = 1000$	$p_Y(y)$
$y = 1000$	0.05	0.10	0.15	0.30
$y = 500$	0.10	0.20	0.05	0.35
$y = 100$	0.20	0.10	0.05	0.35
$p_X(x)$	0.35	0.40	0.25	

- (a) Compute the probabilities $\mathbb{P}(Y = 500|X = 500)$, $\mathbb{P}(Y = 100|X = 500)$.
- (b) Compute the conditional means $\mathbb{E}[X|Y = 100]$, $\mathbb{E}[Y|X = 500]$.

(b)

$$\begin{aligned}\mathbb{E}[X|Y = 100] &= 100 \cdot \frac{p(100, 100)}{p_Y(100)} + 500 \cdot \frac{p(500, 100)}{p_Y(100)} + 1000 \cdot \frac{p(1000, 100)}{p_Y(100)} \\ &= 100 \cdot \frac{0.20}{0.35} + 500 \cdot \frac{0.10}{0.35} + 1000 \cdot \frac{0.05}{0.35} \\ &= \frac{2400}{7}.\end{aligned}$$

$$\begin{aligned}\mathbb{E}[Y|X = 500] &= 100 \cdot \frac{p(500, 100)}{p_X(500)} + 500 \cdot \frac{p(500, 500)}{p_X(500)} + 1000 \cdot \frac{p(500, 1000)}{p_X(500)} \\ &= 100 \cdot \frac{0.10}{0.40} + 500 \cdot \frac{0.20}{0.40} + 1000 \cdot \frac{0.10}{0.40} \\ &= 525.\end{aligned}$$

Problem 3 (Joint PMF, Marginal PMF, Conditional PMF)

Let X and Y have a uniform distribution on the set of points with **integer** coordinates in $S = \{(x, y) : 0 \leq x \leq 7, x \leq y \leq x + 2\}$. That is, $p(x, y) = 1/24$, $(x, y) \in S$, and both x and y are integers. Find

(a) the marginal PMF $p_X(x)$ and $p_Y(y)$.

(b) the conditional PMF $h(y|x)$ of Y given $X = x$.

(c) $\mathbb{E}[Y|X = x]$. (d)

(d) $\sigma_{Y|x}^2$.

$$\begin{aligned}\sigma_{Y|x}^2 &= \mathbb{E}[Y^2|X = x] - \mu_{Y|x}^2 \\ &= x^2 h(x|x) + (x+1)^2 h(x+1|x) + (x+2)^2 h(x+2|x) - (1+x)^2 \\ &= \frac{x^2}{3} + \frac{(x+1)^2}{3} + \frac{(x+2)^2}{3} - (1+x)^2 \\ &= \frac{2}{3},\end{aligned}$$

for $x = 0, 1, 2, 3, 4, 5, 6, 7$.

Problem 4 (Joint PMF, Marginal PMF, Conditional PMF)

Let $p_X(x) = 1/10$, $x = 0, 1, 2, \dots, 9$, and let the conditional PMF of Y given $X = x$ be $h(y|x) = 1/(10 - x)$, $y = x, x + 1, \dots, 9$. Find

- (a) $p(x, y)$.
- (b) $p_Y(3)$.
- (c) $\mathbb{E}[Y|X = 7]$.

(b)

	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$y = 4$	$y = 5$	$y = 6$	$y = 7$	$y = 8$	$y = 9$
$x = 0$	$1/100$	$1/100$	$1/100$	$1/100$	$1/100$	$1/100$	$1/100$	$1/100$	$1/100$	$1/100$
$x = 1$		$1/90$	$1/90$	$1/90$	$1/90$	$1/90$	$1/90$	$1/90$	$1/90$	$1/90$
$x = 2$			$1/80$	$1/80$	$1/80$	$1/80$	$1/80$	$1/80$	$1/80$	$1/80$
$x = 3$				$1/70$	$1/70$	$1/70$	$1/70$	$1/70$	$1/70$	$1/70$
$x = 4$					$1/60$	$1/60$	$1/60$	$1/60$	$1/60$	$1/60$
$x = 5$						$1/50$	$1/50$	$1/50$	$1/50$	$1/50$
$x = 6$							$1/40$	$1/40$	$1/40$	$1/40$
$x = 7$								$1/30$	$1/30$	$1/30$
$x = 8$									$1/20$	$1/20$
$x = 9$										$1/10$

$$\begin{aligned} p_Y(3) &= \sum_{x=0}^3 p(x, 3) \\ &= p(0, 3) + p(1, 3) + p(2, 3) + p(3, 3) \\ &= \frac{1}{100} + \frac{1}{90} + \frac{1}{80} + \frac{1}{70} \approx 0.0479. \end{aligned}$$

(c)

$$\begin{aligned}\mathbb{E}[Y|X = 7] &= \sum_{y=7}^9 yh(y|7) \\ &= 7h(7|7) + 8h(8|7) + 9h(9|7) \\ &= 7(1/3) + 8(1/3) + 9(1/3) \\ &= 8.\end{aligned}$$

Problem 5 (Joint PMF, Marginal PMF, Conditional PMF)

From a standard poker deck of 52 cards, 3 cards are drawn. Let X be number of clubs among the 3 cards and let Y be the number of hearts among the 3.

- (a) Find the joint PMF of X and Y .
- (b) Find the marginal PMFs of X and Y .
- (c) Compute $P(X = 1|Y = 1)$.
- (d) Let F be the joint CDF of X and Y . Compute $F(1, 1)$.

$x \setminus y$	0	1	2	3
0	$\frac{200}{1700}$	$\frac{325}{1700}$	$\frac{156}{1700}$	$\frac{22}{1700}$
1	$\frac{325}{1700}$	$\frac{338}{1700}$	$\frac{78}{1700}$	0
2	$\frac{156}{1700}$	$\frac{78}{1700}$	0	0
3	$\frac{22}{1700}$	0	0	0

(c) We compute

$$\begin{aligned} P(X = 1|Y = 1) &= \frac{P(X = 1, Y = 1)}{P(Y = 1)} \\ &= \frac{p(1, 1)}{p_Y(1)} = \frac{\frac{338}{1700}}{\frac{741}{1700}} = \frac{338}{741}. \end{aligned}$$

(d) We have

$$\begin{aligned} F(1, 1) &= P(X \leq 1, Y \leq 1) \\ &= p(0, 0) + p(0, 1) + p(1, 0) + p(1, 1) \\ &= \frac{200}{1700} + \frac{325}{1700} + \frac{325}{1700} + \frac{338}{1700} = \frac{1188}{1700}. \end{aligned}$$